The KSL12 loudspeaker

The KSL12 is a line array module specifically designed for medium to large-scale sound reinforcement.
Up to twenty-four KSL12 loudspeakers can be flown in vertical columns producing an 120° constant directivity dispersion pattern in the horizontal plane.
The KSL12 houses two 10" neodymium forward facing LF drivers and two side firing 8" neodymium LF drivers. A coaxial mid-high section contains an MF horn with an 8" driver and two 1.4" exit, 3" voice coil HF compression drivers mounted to a dedicated wave shaping device.
Splay angles between cabinets can range from 0° to 10° in 1° increments.
The loudspeakers are driven actively by two channels of an appropriate d&b amplifier, one channel powering the 10" LF drivers, the second channel powers all other components, these are passively crossed-over.
This component geometry allows for a smooth crossover design with well-defined overlaps between adjacent bands providing consistent, even and very accurate horizontal dispersion.
Due to the arrangement of the front and side firing LF drivers, in combination with approprate processing, constant directivity control is maintained from 54 Hz to above 18 kHz.
The cabinet is constructed from marine plywood and have an impact and weather protected PCP (Polyurea Cabinet Protection) finish. The front and side panels incorporate rigid metal grills backed by an acoustically transparent and water repellent fabric. Each side panel incorporates a recessed handle, with additional handles provided at the rear.

d&b amplifiers

The d&b audiotechnik loudspeaker range is designed exclusively for operation with d&b amplifiers. These provide power as well as comprehensive control and protection functions tailored to achieve the performance, reliability and longevity associated with the d&b system approach.

System data

Frequency response (–5 dB standard)	54 Hz - 18 kHz
Frequency response (–5 dB CUT mode)	75 Hz - 18 kHz
Frequency response (–10 dB standard, IEC60268)	48 Hz - 19 kHz
Frequency response (–10 dB CUT mode, IEC60268)	63 Hz - 19 kHz
Max. sound pressure (1 m, free field, broadband signal IEC60268)	
With D90/D80/D40/40D	144 dB
4x KSL12 (far field, scaled to 1 m)	
with D90/D80/D40/40D	156 dB

Loudspeaker data

Nominal impedance front LF	8 ohms
Nominal impedance side LF/MF/HF	8 ohms
Power handling capacity front LF (RMS/peak 10 ms)	
	450/1800 W
Power handling capacity side LF/MF/HF (RMS/peak 10 ms)	
	250/1000 W
Nominal dispersion angle (horizontal)	120°
Splay angle setting	0 ... 10° (1° increment)
Components	2 x 10" front LF driver
	2 x 8" side LF driver
	(2x 12.2" total equivalent* LF driver size)
	1 x 8" MF driver
	2 x 1.4" exit compression driver with 3" coil
	Passive crossover network
Connections	NLT4 F/M
Pin assignment	1+: Front LF+/1–: Front LF–
	2+: Side LF/MF/HF+/2–: Side LF/MF/HF–
Weight	58 kg (128 lb)

* Calculated equivalent diameter of coherent addition of front and side radiating LF drivers.

d&b audiotechnik

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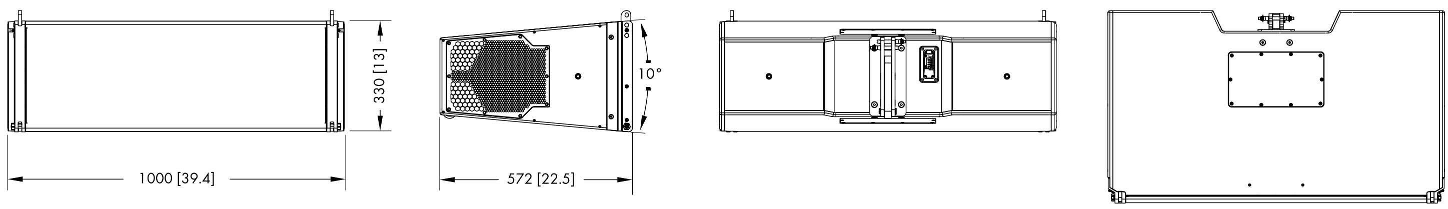

KSL12 cabinet dimensions in mm [inch]

Features and benefits

- Constant directivity behavior over the entire operating range using cardioid techniques in the lower range
- Exceptional broadband headroom
- Patented SL-Series rigging hardware and method enables rapid deployment of arrays directly from a touring cart in either compression or tension rigging modes
- Requires only two amplifier channels; one channel drives the front facing LF drivers, while the other amplifier channel drives the passively crossed over side firing LF drivers, the MF section and two HF drivers
- ArrayProcessing optimizes the level and tonal balance over the complete audience listening area
- For short arrays where ArrayProcessing is not required, two loudspeakers can be linked and driven in the Line/Arc mode
- Efficient cabling system and amplifier rack assemblies
- Effective transport solutions

Applications

- Medium and large scale sound reinforcement applications
- Stadiums and arenas
- Concert halls
- Houses of Worship
- Theaters
- Clubs and live music venues

Architectural specifications

The loudspeaker system shall consist of two forward 10" LF neodymium drivers in a vented enclosure radiating to the front, two sideward 8" LF neodymium drivers, one hornloaded 8" midrange driver and two coaxially mounted 1.4" exit compression drivers with 3″ voicecoils coupled to a waveshaping device.
The loudspeaker system shall be 3-way, actively driven between the forward LF drivers and the sideward LF driver with mid/high sections. Passive crossovers shall be used between the sideward LF driver and the mid/high sections.
The total equivalent front facing LF driver size shall be 2x 12.2".
The loudspeaker shall only be operated by a dedicated, compatible controller amplifier.
The enclosure shall be constructed from marine plywood with an impact resistant and weather protecting PCP (Polyurea Cabinet Protection) finish.
The cabinet front and side shall be protected by a perforated steel grill backed with acoustically transparent and water repellent fabric.
The cabinet shall incorporate a three point rigging system for the assembly of vertical line source arrays of up to 24 cabinets in connection with a dedicated flying frame.
The power handling of the forward LF section shall be 450/1800 W while the power handling of the sideward LF drivers and MF/HF section shall be 250/1000 W (RMS/peak 10 ms).
The frequency response (–5 dB) measured on axis shall extend from 54 Hz - 18 kHz.
Maximum sound pressure level (SPLmax peak/1 m/free field) of single cabinet shall be at least 144 dB while maximum sound pressure level (SPLmax far field, scaled to1 m) of 4 modules shall be at least 156 dB.
The horizontal dispersion shall be 120°, while the vertical splay angle shall be adjustable in a range of 0° - 10° in 1° increments.
The connection panel on the back shall be recessed and fitted with speakON NLT4 F/M sockets.
The dimensions (W x H x D) shall not exceed 1000 x 330 x 597 mm (39.4" x 13" x 23.5") and shall weigh no more than 58 kg (128 lb).

The loudspeaker shall be the KSL12 by:
d&b audiotechnik GmbH & Co. KG.